

Response to reviewer for

The biodiversity we ignore: a global meta-analysis on taxonomic bias

Editor:

As 10 reviewers declined the invitation to review the manuscript, and since this is a resubmission, I have decided to follow the recommendation of the sole reviewer and request a minor revision.

**Kind regards,
Angela Wul**

RESPONSE:

Dear Editor,

The reviewer market is indeed becoming increasingly challenging – we really appreciate your effort and perseverance in finding reviewers. We have carefully addressed all reviewer comments and believe the manuscript is substantially improved. See below for details (and we have marked main changes to the text in red).

Reviewer #1:

My only main concern is that I wonder why didn't you include "Speciesism" and "taxonomic bias" in the bias part of the literature search. I would add these terms as well, redo the search, and eventually incorporate any new studies that may appear (it can be the case that it does not provide any additional study).

RESPONSE: We thank the reviewer for this valuable suggestion. We did not originally include the term taxonomic bias because it describes the phenomenon under investigation rather than a specific theoretical framework or mechanism of bias (e.g., plant blindness or vertebratism). Conversely, we became aware of the concept of speciesism while reviewing the literature retrieved for this meta-analysis, suggesting that our broader search terms were already capturing at least some studies using this terminology.

To assess the potential impact of this omission, we repeated the search including the term speciesism. This yielded six additional records; however, none met our inclusion criteria and/or reported extractable statistics suitable for inclusion in the meta-analysis. Consequently, the dataset and results remained unchanged.

Given that the inclusion of speciesism did not alter the final set of studies, and acknowledging that no search strategy can capture every possible term describing biodiversity-related biases, we chose not to modify the search protocol retrospectively. Instead, we clarified in the Methods that our objective was to obtain a representative and meaningful sample of studies investigating disparities in biodiversity knowledge, rather than an exhaustive census of all potentially relevant publications:

“We systematically collected *a sample of studies* investigating disparities in biodiversity knowledge using the Web of Science database...”

GENERAL COMMENTS:

I think this is a great paper. I only have a significant concern about the search, which may require to do additional analyses. But it is unlikely that it changes the results in a qualitative way. Note, however, that although I've done some meta-analyses myself I'm not an expert, so other researchers may have different criticisms. In any case, I congratulate you for help the field pointing the finger to biases.

RESPONSE: Thank you for the positive feedback!

SPECIFIC COMMENTS:

Besides that I have a number of specific comments:

Keywords: I think the right Word is Speciesism

Also, I would add "conservation bias" to the keywords

RESPONSE: done

Lines 34-6. I tend to not like questions written as questions (I mean, with question marks) in scientific papers, except when they are strictly necessary. I understand it can be justified that sometimes the research questions addressed by the paper are written that way. But in this case I think it would be better to write these sentences in a different way, for example "(...) leaving ample room for debate on whether we have enough knowledge to determine if the ongoing biodiversity decline can be described as a sixth mass extinction (REF), or which branches of the tree of life are most affected by these extinctions (REFS)."

RESPONSE: done

Line 38-9. Avoid wordy statements; you can simply start the sentence with "Research in ecology, behaviour (...)". Note also that the references you cite are from 2002 and not 2022.

Line 40. Related to the former comment, I don't think these papers were "early"; taxonomic bias was already under debate at least in the 1990s, if not earlier; see e.g. Metrick & Weitzman (Land Economics 1996) or Moyle (Ecological Economics 1998). I would just delete "early" here

RESPONSE: Rather than removing early (which connected to the historical account later on), we rephrased this part to better account for the landscape of “early” studies, including the two suggested references, as follow: *‘In early contributions, it was shown that research in ecology, behaviour, and conservation is dominated by studies on birds and mammals, while less ‘popular’ groups like amphibians and reptiles receive far less attention – phenomena described as “taxonomic chauvinism” (Bonnet et al. 2002) or “taxonomic bias” (Clark & May 2002). These early insights and others (e.g., Metrick & Weitzman 1996; Moyle 1998) catalyzed a broad research agenda aimed at documenting and quantifying biases in biodiversity knowledge. Researchers soon recognized that this bias, ... ’*

Lines 54-5. Contrary to what I said earlier, here the questions work well in the format they are written.

RESPONSE: perfect!

Line 62. Add "totalling", "encompassing", "including" or something similar before "905"

RESPONSE: Done

Line 128. Add "of" after awareness

RESPONSE: Done

Line 138. Add a reference for the use of inter-rater agreement tests

RESPONSE: We cited Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educ. Psychol. Meas.*, 20(1), 37–46.

Lines 251-2. You quote an interval around the average number of effect sizes with breadth values higher than the average, which would lead to negative values in the lower bound - which is obviously impossible. This is a good indication that you shouldn't be using averages, but rather median values for the central trend of effect sizes per study, and the minimum and maximum values for the range of values (or the interquantile range).

RESPONSE: We now reported median and range.

Line 278. "variable" in singular

RESPONSE: done

Lines 373-9. I would also mention in this paragraph the bias in the meta-literature you started the discussion with. I think the kick-off paragraph of the discussion is great, so I won't change it or move anything to other place. But here you could also mention that bias, perhaps saying that as new studies accumulate, if they keep biased towards vertebrates, evidence on neglected groups will continue to be limited or inexistent.

RESPONSE: Great point, we expanded the paragraph as follows: *‘A poignant example of this is the bias in the meta-literature itself we documented here, with vertebrates dominating studies on bias. It is likely that birds and mammals benefit from greater availability of naturalistic observations, distribution data, genetic data, trait data, and cultural narratives. This data richness allows for far greater and more detailed research on the taxonomic bias affecting these groups. As new studies accumulate, it will be possible to keep expanding our meta-analysis database and obtain stronger inference over time – but only if the taxonomic and geographic breadth of new studies also increases.’*

I don't think Figure 4 works particularly well in this context. It may be good for

150 **outreach or to present this research in talks and conferences, but here in the**
151 **middle of the paper is reductionist and unnecessary. I would delete it.**
152 RESPONSE: done as suggested. We repurposed it as a graphical abstract.